
Golden-Section Search

Derivative-free optimization by interval reduction

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Four questions

- 1 Why does the constant $0.618\dots$ appear?
- 2 How can one function value be reused at every step?
- 3 What accuracy is guaranteed after a fixed budget?
- 4 When should we *not* trust the answer?

The working viewpoint

The method does not chase a derivative. It shrinks a bracket that is known to contain one optimum.

Problem Setting

1

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A derivative-free question

$$x^* = \arg \max_{x \in [x_L, x_U]} f(x)$$

Only request: values of $f(x)$.

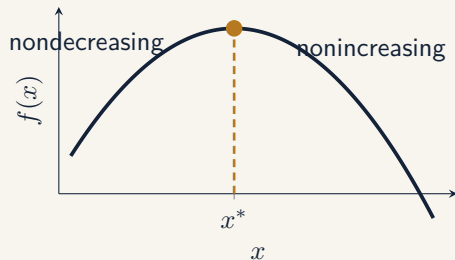
A reliable derivative may be unavailable, expensive, or meaningless.

Typical situations

- ▶ laboratory measurements;
- ▶ simulation runs;
- ▶ noisy or tabulated output.

For minimization, replace $\arg \max$ by $\arg \min$; the geometry is unchanged.

The unimodality contract



A maximum at x^* divides the bracket:

- ▶ f is nondecreasing up to x^* ;
- ▶ f is nonincreasing after x^* .

What is not required

No derivative, quadratic model, or smooth graph is required.

The bracket is a claim

- 1 **Valid domain:** every trial point is safe to evaluate.
- 2 **Contains the target:** the desired optimum is inside.
- 3 **One relevant mode:** one peak or trough governs the interval.

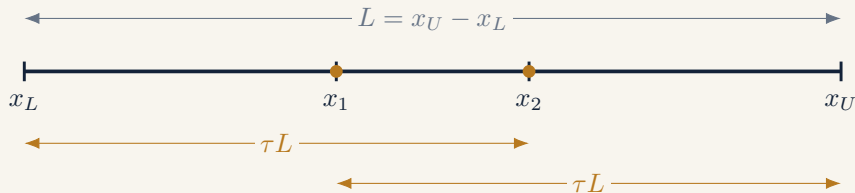
Why this matters

The method certifies a smaller *interval*; it cannot repair a bracket that excluded the optimum or mixed several extrema.

The Search Mechanism

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Two probes, one bracket



$$x_1 = x_U - \tau L$$

$$x_2 = x_L + \tau L$$

$$\tau = \frac{\sqrt{5} - 1}{2} \approx 0.618034, \quad x_L < x_1 < x_2 < x_U.$$

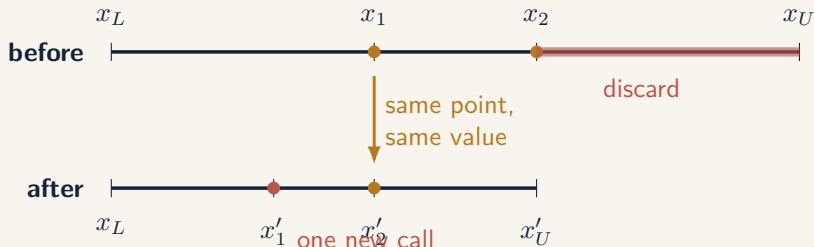
Which side survives?

Comparison	Maximization keeps	Minimization keeps
$f(x_1) > f(x_2)$	$[x_L, x_2]$	$[x_1, x_U]$
$f(x_1) < f(x_2)$	$[x_1, x_U]$	$[x_L, x_2]$
$f(x_1) = f(x_2)$	$[x_1, x_2]$	$[x_1, x_2]$

Read one row, not the whole table

For a maximum, the larger value points toward the survivor. For a minimum, the smaller value does.

Reuse one evaluation



For the illustrated maximization case, $f(x_1) > f(x_2)$. Therefore

$$x'_U \leftarrow x_2, \quad x'_2 \leftarrow x_1, \quad f(x'_2) \leftarrow f(x_1).$$

Only x'_1 is new, so only **one new function call** is needed.

Algorithm: initialize once

- 1 Set $\tau = (\sqrt{5} - 1)/2$ and $L = x_U - x_L$.
- 2 Form $x_1 = x_U - \tau L$ and $x_2 = x_L + \tau L$.
- 3 Evaluate $f_1 = f(x_1)$ and $f_2 = f(x_2)$.

State carried into the loop

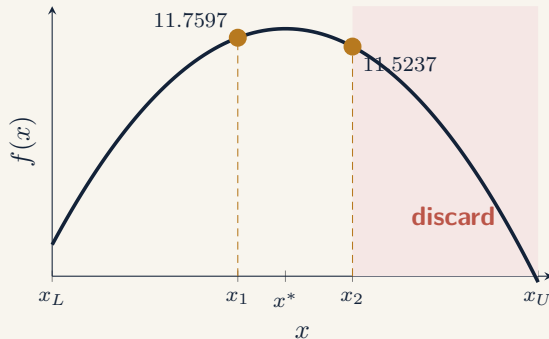
Keep the ordered endpoints, two probes, and their two function values. Losing a saved value also loses the main efficiency advantage.

Algorithm: repeat

- 1 Compare f_1 and f_2 .
- 2 Discard the side ruled out by unimodality.
- 3 Rename the surviving probe and reuse its value.
- 4 Create one new probe and evaluate f there.
- 5 Stop when the bracket meets the accuracy rule.

For minimization, reverse only the two strict comparison decisions; the probe geometry and reuse are unchanged.

Worked trace: first cut



Maximize

$$f(x) = 12 - (x - 2.4)^2$$

on $[0, 5]$.

$$x_1 = 1.90983, \quad x_2 = 3.09017.$$

Since

$$f(x_1) > f(x_2),$$

retain **$[0, 3.09017]$** .

Worked trace: cuts 0-2

k	Bracket before	x_1	x_2	Retained bracket
0	$[0, 5]$	1.90983	3.09017	$[0, 3.09017]$
1	$[0, 3.09017]$	1.18034	1.90983	$[1.18034, 3.09017]$
2	$[1.18034, 3.09017]$	1.90983	2.36068	$[1.90983, 3.09017]$

Watch the gold entry

It moves to the other probe column because the physical point and its function value are reused.

Worked trace: cuts 3-5

k	Bracket before	x_1	x_2	Retained bracket
3	[1.90983, 3.09017]	2.36068	2.63932	[1.90983, 2.63932]
4	[1.90983, 2.63932]	2.18847	2.36068	[2.18847, 2.63932]
5	[2.18847, 2.63932]	2.36068	2.46711	[2.18847, 2.46711]

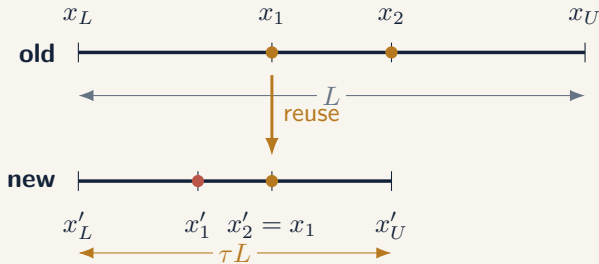
Result after six comparisons

The true maximizer $x^* = 2.4$ remains inside $[2.18847, 2.46711]$, now only 0.27864 wide.

Convergence and Cost

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Why does 0.618 appear?



The old x_1 is $(1 - \tau)L$ from the left endpoint.

In the new bracket, the right probe is $\tau(\tau L) = \tau^2 L$ from that endpoint. Reuse requires

$$1 - \tau = \tau^2.$$

Hence

$$\tau = \frac{\sqrt{5} - 1}{2}.$$

Why the discard is safe

Consider maximization with $x_1 < x_2$ and $f(x_1) > f(x_2)$.

- 1 Suppose, for contradiction, that $x^* > x_2$.
- 2 Both probes would then lie on the nondecreasing side of the peak.
- 3 Unimodality would require $f(x_1) \leq f(x_2)$, a contradiction.

Therefore $x^* \leq x_2$, so discarding $(x_2, x_U]$ preserves the invariant

$$x^* \in [x_L, x_U].$$

The other comparison and minimization cases are symmetric.

Rate and error guarantee

Exact contraction

$$L_k = \tau^k L_0.$$

For midpoint output,

$$|\hat{x}_k - x^*| \leq \frac{L_k}{2}.$$

Required reductions

$$k \geq \max \left\{ 0, \left\lceil \frac{\log(2\varepsilon/L_0)}{\log \tau} \right\rceil \right\}.$$

Example

If $L_0 = 5$ and $\varepsilon = 10^{-3}$, then $k = 17$. The actual bound is about 7.00×10^{-4} .

What actually converges?

Bracket width: exactly Q-linear,

$$\frac{L_{k+1}}{L_k} = \tau.$$

Location estimate: its guaranteed error is at most $L_k/2$, also decreasing linearly.

Objective value: unimodality alone gives no numerical rate. Extra regularity is needed. For example, if f is M -Lipschitz,

$$0 \leq f(x^*) - f(\hat{x}_k) \leq \frac{ML_k}{2}$$

for maximization. A smooth stationary optimum can yield an $O(L_k^2)$ objective error instead.

Count calls, not loops

Initialization

2 function calls

produce the first comparison.

With a stop-after-update implementation,

Each later comparison

1 new function call

because one value is reused.

$$n \text{ comparisons} \implies n + 1 \text{ evaluations.}$$

For an evaluation budget $E \geq 2$, the worst-case bracket is

$$L_{E-1} = \tau^{E-1} L_0.$$

This is why evaluation count is the meaningful complexity measure when f is a simulation or experiment.

Fixed-rule alternatives

- 1 **Equal-thirds search:** two new calls per comparison; retain $2/3$ of the bracket.
- 2 **Fibonacci search:** finite-budget minimax optimal; changing ratios require choosing the budget first.

Golden section's compromise: one stationary ratio, one reused value, and no predetermined final budget.

Reference: Kiefer, "Sequential minimax search for a maximum," Proc. AMS 4 (1953), 502-506.

Brent's hybrid

- ▶ Propose an inverse parabolic interpolation step from sampled values.
- ▶ Accept it when safe; otherwise take a golden-section step.

Trade-off: golden section gives transparent worst-case contraction. Brent adds bookkeeping to gain speed when the curve is informative.

Reference: SciPy 1.18 `minimize_scalar` documentation; Brent (1973).

Reliable Computation



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Stop safely near zero

A common relative test contains $x_U + x_L$ in the denominator. On a symmetric bracket such as $[-1, 1]$, it divides by zero.

For midpoint output, use the scale-aware condition

$$L_k \leq 2 (\varepsilon_{\text{abs}} + \varepsilon_{\text{rel}} \max\{|x_L|, |x_U|\}).$$

It directly guarantees

$$|\hat{x}_k - x^*| \leq \varepsilon_{\text{abs}} + \varepsilon_{\text{rel}} \max\{|x_L|, |x_U|\}.$$

Also stop on an evaluation budget or when rounding creates no distinct probe.

Implementation checks

Preserve state

- ▶ move each (x_i, f_i) pair together;
- ▶ cache the surviving evaluation;
- ▶ assert

$$x_L < x_1 < x_2 < x_U.$$

Once machine arithmetic cannot create a distinct probe, further interval algebra cannot create new information.

Exit honestly

- ▶ reject NaN or Inf values;
- ▶ stop if a probe rounds to an endpoint;
- ▶ return bracket, status, and call count.

How the method misleads

Several extrema

A legal-looking comparison can discard the globally best basin.

Noisy evaluations

One flipped comparison causes an irreversible wrong cut.

Flat optimum

The bracket shrinks, but the optimizer need not be unique.

Invalid values

NaN, Inf, or domain errors destroy the ordering decision.

Wrong inequality

A minimization update used for maximization walks away from the peak.

False precision

A tiny bracket does not correct model error or measurement bias.

Modern adaptations

Line search in many dimensions

$$\phi(\alpha) = F(\mathbf{x} + \alpha \mathbf{d})$$

turns one step-length choice into a scalar golden-section problem.

Noisy or expensive objectives

- ▶ cache results, repeat uncertain comparisons, and enforce a budget;
- ▶ stop when noise exceeds the improvement being resolved.

The method is sequential: one new call per step saves evaluations, but the next location must wait. Parallel hardware may favor a batched alternative.

Mathematical Problems

5

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Problem 1: Golden ratio

Normalize a search bracket to $[0, 1]$. Place the two probes at

$$x_1 = 1 - \tau, \quad x_2 = \tau, \quad \frac{1}{2} < \tau < 1.$$

Suppose the right-hand bracket $[x_1, 1]$ survives. Choose τ so that the old point x_2 is exactly the left probe of the new bracket.

- 1 Form the self-similarity equation for τ .
- 2 Solve it and reject the inadmissible root.
- 3 State the fraction of length removed by one reduction.

No decimal approximation is needed until the last line.

Solution 1: Derivation

The surviving bracket has lower end $1 - \tau$ and length τ . Its left probe is therefore

$$(1 - \tau) + (1 - \tau)\tau = 1 - \tau^2.$$

Requiring this to coincide with the old point $x_2 = \tau$ gives

$$1 - \tau^2 = \tau \iff \tau^2 + \tau - 1 = 0.$$

Hence

$$\tau = \frac{-1 \pm \sqrt{5}}{2}.$$

The negative root is outside $(1/2, 1)$, so

$$\tau = \frac{\sqrt{5} - 1}{2} \approx 0.618034.$$

The retained fraction is τ ; the discarded fraction is $1 - \tau = \tau^2 \approx 0.381966$.

Problem 2: Hand trace

Use golden-section search to minimize

$$f(x) = (x - 1.3)^2 + 2, \quad 0 \leq x \leq 4.$$

With $\tau = (\sqrt{5} - 1)/2$, carry out the first **four** bracket reductions. At each step, show

$$x_1 = x_U - \tau(x_U - x_L), \quad x_2 = x_L + \tau(x_U - x_L),$$

the two function values, and the retained bracket.

Finally, report the midpoint estimate and the number of function evaluations used.

Solution 2: Trace table

For minimization, $f(x_1) < f(x_2)$ keeps $[x_L, x_2]$; otherwise it keeps $[x_1, x_U]$.

k	x_1	x_2	$f(x_1)$	$f(x_2)$	retained bracket
1	1.5279	2.4721	2.0519	3.3739	$[0, 2.4721]$
2	0.9443	1.5279	2.1265	2.0519	$[0.9443, 2.4721]$
3	1.5279	1.8885	2.0519	2.3464	$[0.9443, 1.8885]$
4	1.3050	1.5279	2.0000	2.0519	$[0.9443, 1.5279]$

Thus

$$\hat{x} = \frac{0.9443 + 1.5279}{2} = \boxed{1.2361}.$$

Two calls initialize the method; each later reduction needs one new call. Four reductions therefore use $2 + 3 = \boxed{5}$ evaluations.

Check: the exact minimizer 1.3 remains inside every retained bracket.

Problem 3: Error budget

A unimodal objective is bracketed on $[2, 18]$. After k golden-section reductions, the midpoint of the surviving bracket will be reported.

Find the smallest integer k that guarantees

$$|\hat{x} - x^*| \leq 5 \times 10^{-4}.$$

Your solution must:

- 1 start from the bracket-length formula;
- 2 verify that one fewer reduction is insufficient;
- 3 convert the result into a function-evaluation budget.

Solution 3: Budget

The initial length is $L_0 = 18 - 2 = 16$, and after k reductions

$$L_k = 16\tau^k, \quad |\hat{x} - x^*| \leq \frac{L_k}{2} = 8\tau^k.$$

Therefore

$$8\tau^k \leq 5 \times 10^{-4} \implies k \geq \frac{\ln(6.25 \times 10^{-5})}{\ln \tau} \approx 20.12.$$

Thus the smallest integer is $\boxed{k = 21}$.

The boundary check is

$$8\tau^{20} \approx 5.289 \times 10^{-4} > 5 \times 10^{-4}, \quad 8\tau^{21} \approx 3.269 \times 10^{-4} < 5 \times 10^{-4}.$$

With cached reuse, k reductions require $k + 1$ calls. The guaranteed budget is therefore $\boxed{22 \text{ function evaluations}}$.

Problem 4: Safe discard

Let f be strictly unimodal with a unique maximum x^* on $[x_L, x_U]$, and let

$$x_L < x_1 < x_2 < x_U.$$

Prove directly from strict unimodality that

$$f(x_1) < f(x_2) \implies x^* \in [x_1, x_U].$$

Then determine what the equality $f(x_1) = f(x_2)$ says about the location of x^* .

Allowed fact

A strictly unimodal function increases before x^* and decreases after x^* .

Solution 4: Proof

Case 1: $f(x_1) < f(x_2)$.

Suppose instead that $x^* < x_1$. Both probes would then lie on the strictly decreasing side, so

$$x^* < x_1 < x_2 \implies f(x_1) > f(x_2),$$

a contradiction. Therefore

$$x^* \in [x_1, x_U].$$

Case 2: $f(x_1) = f(x_2)$.

If $x^* \leq x_1$, strict decrease gives $f(x_1) > f(x_2)$. If $x^* \geq x_2$, strict increase gives $f(x_1) < f(x_2)$. Both are impossible, hence

$$x^* \in (x_1, x_2).$$

A flat-topped objective needs a weaker conclusion about its set of maximizers.

Problem 5: Stopping rule

Consider the relative-width test

$$2 \left| \frac{x_U - x_L}{x_U + x_L} \right| < \delta.$$

It is applied to $[x_L, x_U] = [-0.002, 0.002]$ with $\delta = 10^{-3}$.

- 1 Explain exactly why the test fails.
- 2 Replace it with a mixed absolute-relative test for the midpoint, using $\varepsilon_{\text{abs}} = 10^{-4}$ and $\varepsilon_{\text{rel}} = 10^{-3}$.
- 3 Find how many additional golden reductions guarantee the new test.

Solution 5: Scale floor

Here $x_U + x_L = 0$, so the original test is undefined even though the bracket length is 0.004. Use

$$\frac{L}{2} \leq \varepsilon_{\text{abs}} + \varepsilon_{\text{rel}} \max(|x_L|, |x_U|).$$

The present length tolerance is 2.04×10^{-4} . Since the relative scale can fall as the bracket moves, a uniform guarantee uses the absolute floor $L \leq 2 \times 10^{-4}$. With $L_m = 0.004\tau^m$,

$$m \geq \frac{\ln(2 \times 10^{-4}/0.004)}{\ln \tau} \approx 6.23, \quad \boxed{m = 7}.$$

Indeed, $L_6 \approx 2.229 \times 10^{-4}$ fails, while $L_7 \approx 1.378 \times 10^{-4}$ passes. The absolute term prevents the denominator pathology near zero.

Application Problems

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Application 1: Damper

A suspension engineer tunes a dimensionless damping setting c . A finite-element run predicts the peak displacement

$$D(c) = 1.5 + 12(c - 0.38)^2 \text{ mm}, \quad 0.10 \leq c \leq 0.90.$$

The engineer treats the simulator as a black box and minimizes D using golden-section search. One run takes 8 minutes.

- 1 Carry out the first three bracket reductions.
- 2 Report the final bracket and its midpoint.
- 3 Find the total simulation time when values are cached.

The formula is supplied so that the simulator outputs can be checked by hand.

Solution 1: Damper result

For minimization, retain the side containing the smaller displacement.

k	c_1	c_2	$D(c_1)$	$D(c_2)$	retained bracket
1	0.4056	0.5944	1.5078	2.0517	$[0.1000, 0.5944]$
2	0.2889	0.4056	1.5997	1.5078	$[0.2889, 0.5944]$
3	0.4056	0.4777	1.5078	1.6146	$[0.2889, 0.4777]$

Hence

$$c \in [0.2889, 0.4777], \quad \hat{c} = \frac{0.2889 + 0.4777}{2} = \boxed{0.3833}.$$

Three reductions need $3 + 1 = 4$ cached evaluations, so the elapsed simulation time is $4(8) = \boxed{32 \text{ minutes}}$.

Audit: the model's exact best setting, $c = 0.38$, is inside the bracket.

Application 2: Insulation

A building model balances insulation cost against heat loss:

$$C(t) = 180t + 100e^{-10t}, \quad 0.02 \leq t \leq 0.32,$$

where t is thickness in metres and C is annual cost in thousands of taka. One energy-model run takes 45 seconds.

Use golden-section search to guarantee a midpoint thickness error of at most 1 mm.

- 1 Determine the required reductions and model calls.
- 2 Convert the call count into elapsed time.
- 3 Report the computed bracket and midpoint thickness.

Solution 2: Result

Here $L_0 = 0.32 - 0.02 = 0.30$ m. The midpoint guarantee requires

$$\frac{0.30\tau^k}{2} \leq 0.001 \implies k \geq \frac{\ln(0.001/0.15)}{\ln \tau} \approx 10.41.$$

Thus $k = 11$ reductions and 12 model calls are needed. The corresponding time is

$$12(45) = 540 \text{ s} = 9 \text{ minutes}.$$

Running the updates gives

$$[t_L, t_U] = [0.170466, 0.171973] \text{ m}, \quad \hat{t} = 0.171220 \text{ m} = 171.2 \text{ mm}.$$

Independent audit: $C'(t) = 0$ gives $t^* = \ln(1000/180)/10 = 0.171480$ m, which lies in the bracket.

Application 3: Dosage

A dose-response surrogate assigns the net clinical benefit

$$B(d) = 95 - 0.12(d - 14)^2, \quad 5 \leq d \leq 25,$$

where d is a dose in milligrams. The approved interval is unimodal, and the benefit must be maximized without using derivatives.

The design requirement is stated in outcome units:

$$B(d^*) - B(\hat{d}) \leq 0.05 \text{ benefit points.}$$

Find the smallest guaranteed reduction count, the evaluation count, and the midpoint recommendation produced by golden-section search.

Solution 3: Benefit bound

After k reductions, the midpoint error obeys

$$|\hat{d} - d^*| \leq 10\tau^k \implies B(d^*) - B(\hat{d}) \leq 0.12(10\tau^k)^2.$$

Requiring this bound to be at most 0.05 gives

$$0.12(10\tau^k)^2 \leq 0.05 \implies k \geq 5.69.$$

Therefore $k = 6$, requiring 7 benefit evaluations.

The boundary check is $0.0976 > 0.05$ at $k = 5$, but $0.0373 < 0.05$ at $k = 6$. Six updates give $[13.3282, 14.4427]$ mg, so

$$\hat{d} = 13.8854 \text{ mg}.$$

Its actual model loss is only 0.0016; 0.0373 is the advance guarantee.

Application 4: Pricing

A retailer tests price p , measured in hundreds of taka. A demand simulator predicts weekly profit, in thousands of taka, as

$$\Pi(p) = 12(p - 8)e^{-0.06p}, \quad 10 \leq p \leq 40.$$

Each simulator call takes 20 minutes.

- 1 Compute the initial golden probes and their profits.
- 2 Perform the first maximization update and identify the reused point.
- 3 For eight reductions, compare cached golden search with an implementation that evaluates two probes every time.

Solution 4: Reuse pays

The initial probes and profits are

$$p_1 = 21.4590, \quad \Pi(p_1) = 44.568, \quad p_2 = 28.5410, \quad \Pi(p_2) = 44.472.$$

Since $\Pi(p_1) > \Pi(p_2)$, maximization retains

$$[10, 28.5410].$$

The old $p_1 = 21.4590$ becomes the new right probe; only the new left probe

$$p'_1 = 28.5410 - \tau(28.5410 - 10) = 17.0820$$

needs evaluation.

For eight reductions,

$$\text{cached calls} = 8 + 1 = 9, \quad \text{two fresh probes each time} = 16.$$

Reuse saves 7 calls, or $7(20) = 140$ minutes.

Application 5: Solar tilt

A solar-field simulation maximizes daily energy over $10^\circ \leq \theta \leq 50^\circ$. The first golden probes are

$$\theta_1 = 25.279^\circ, \quad \theta_2 = 34.721^\circ.$$

Weather variability produces these 95% intervals, in kWh:

$$\theta_1 : 5.84 \pm 0.06, \quad \theta_2 : 5.88 \pm 0.06.$$

After replication they narrow to $[5.82, 5.86]$ and $[5.87, 5.91]$.

Explain the correct decision before and after replication, find the next new probe, and calculate the cost of five runs per probe at two minutes per run.

Solution 5

Initially the intervals overlap, so the sign of $E(\theta_2) - E(\theta_1)$ is unresolved. A standard golden-section discard would risk deleting the true maximum.

After replication,

$$[5.82, 5.86] \cap [5.87, 5.91] = \emptyset,$$

and the second probe is reliably better. For maximization, retain

$$\boxed{[25.279^\circ, 50^\circ]}.$$

The old 34.721° point becomes the new left probe. The new right probe is

$$25.279 + \tau(50 - 25.279) = \boxed{40.557^\circ}.$$

Five runs at each original probe require 10 runs, costing $10(2) = \boxed{20 \text{ minutes}}$. Later comparisons should also replicate only when their uncertainty intervals overlap.

Summary & References

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